

## Multiple Choice (10 marks)

1. Mars might be a place for future human explorations. However humans can not breathe on the surface of Mars because the atmosphere consists mostly of ...
  - (a) Nitrogen
  - (b) Argon
  - (c) Methane
  - (d) CO<sub>2</sub>
2. What would weigh the most on the moon?
  - (a) A kilogram of feathers
  - (b) Five pounds of bricks as measured on Earth
  - (c) Five kilograms of feathers
  - (d) A kilogram of bricks
3. You've made a scientific theory that there is an attractive force between all objects. When will your theory be proven to be correct?
  - (a) The first time you drop a bowling ball and it falls to the ground proving your hypothesis.
  - (b) After you've repeated your experiment many times.
  - (c) You can never prove your theory to be correct only "yet to be proven wrong".
  - (d) When you and many others have tested the hypothesis.
4. One astronomical unit (AU) is equal to approximately ...
  - (a) 130 million km
  - (b) 150 million km
  - (c) 170 million km
  - (d) 190 million km
5. The visible part of the electromagnetic spectrum is between ...
  - (a) 240 to 680 nm.
  - (b) 360 to 620 nm.
  - (c) 380 to 740 nm.
  - (d) 420 to 810 nm.

## True / False (10 marks)

*Answer True or False*

6. True or False: The Chicxulub Crater, located on the Yucatan Peninsula in Mexico, is linked to the mass extinction event that eliminated the dinosaurs 65 million years ago.
7. True or False: The U.S.S.R. was the first country to successfully send landers to the surface of Venus during the Venera program.
8. True or False: Titan is the only moon in the outer solar system known to possess a thick atmosphere primarily composed of hydrocarbons.
9. True or False: The ratio of solar radiation flux on Mercury's surface at perihelion to aphelion is 4:1.
10. True or False: A lunar day, defined as the time from one sunrise to the next, lasts approximately 29 Earth days.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Discuss the essential properties of our solar system that formation theories must address, and critically analyze why the presence of life on Earth is not a necessary component of these theories.
12. Discuss the methods used to determine the distance of the rock identified as "bigfoot" on Mars, given its size and angular measurement. How do these calculations enhance our understanding of Martian terrain?

*End of Paper*